

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: IFU-Rev-Ch.sil etc.

Creation date: 9-Apr-81 17:02:35

For: Spruce

25 total sheets = 24 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

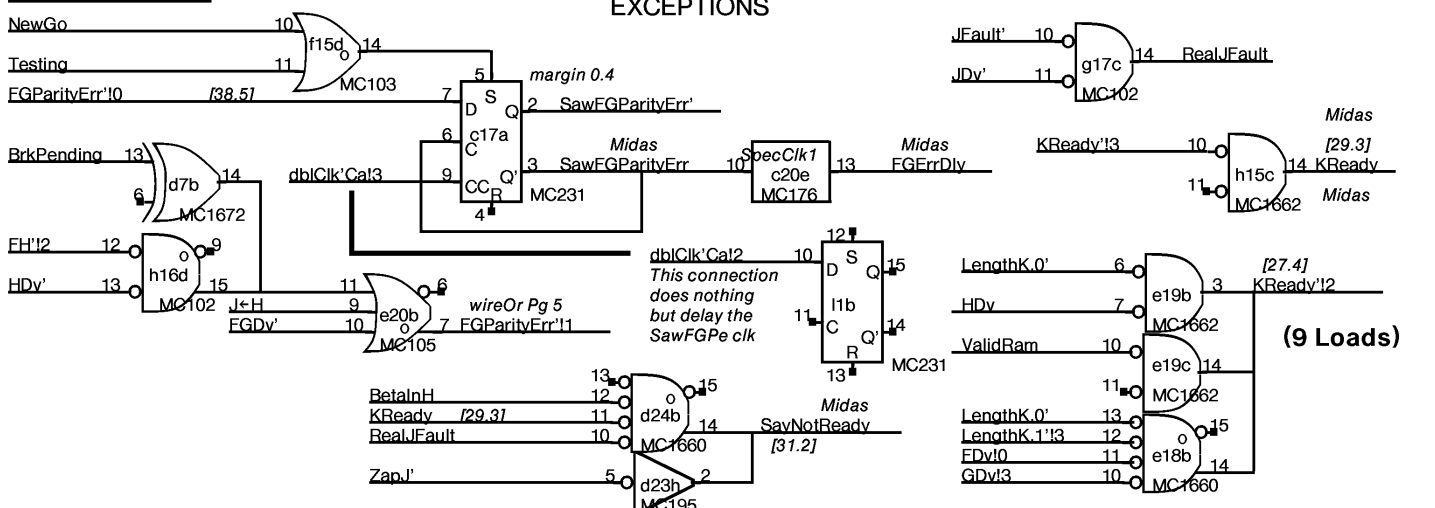
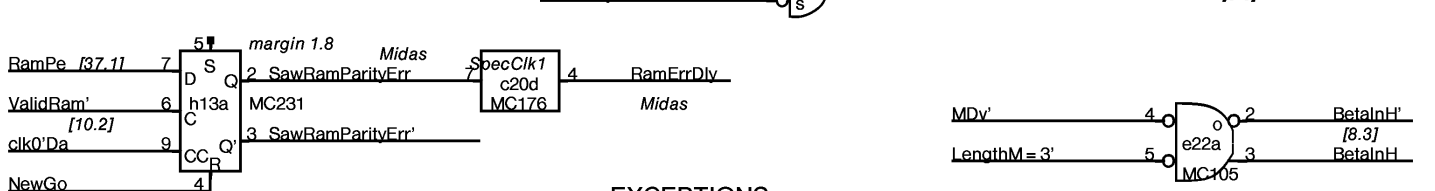
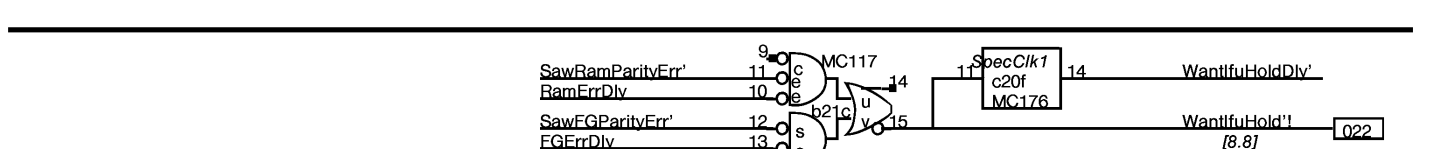
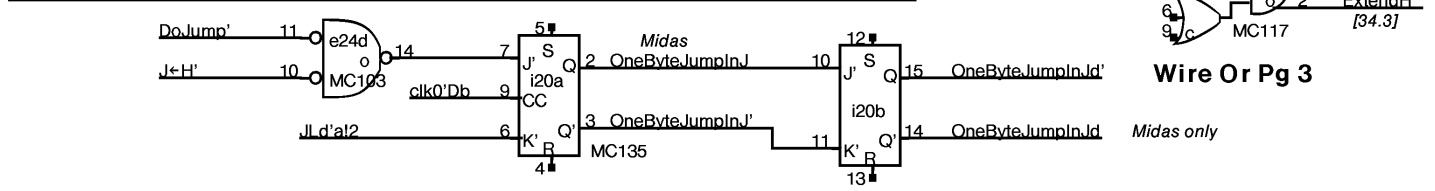
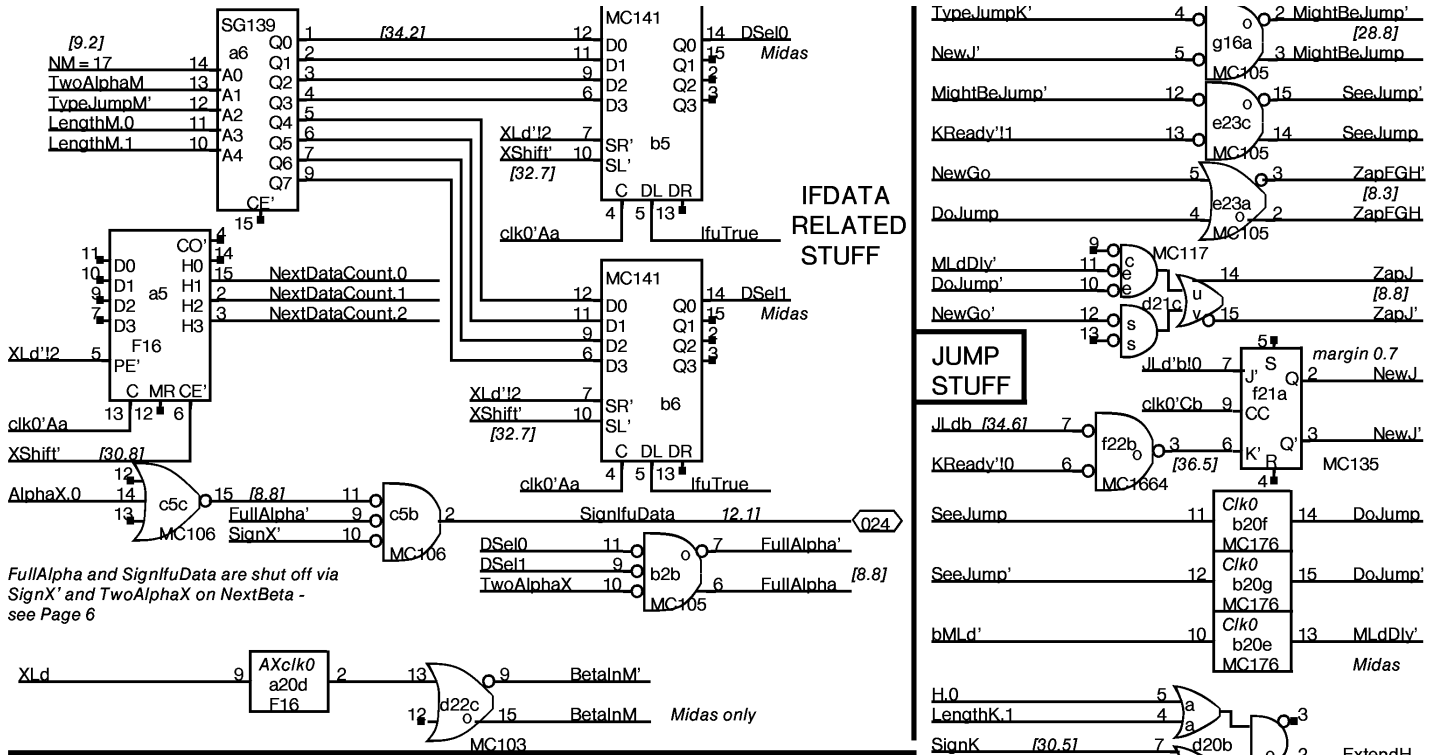
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DORADO SCHEMATICS

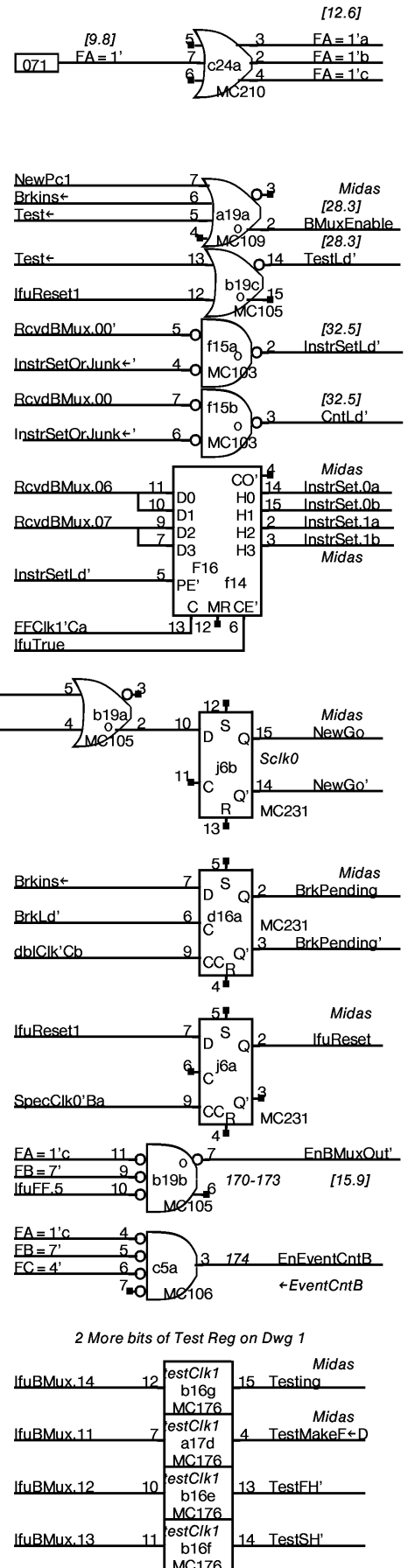
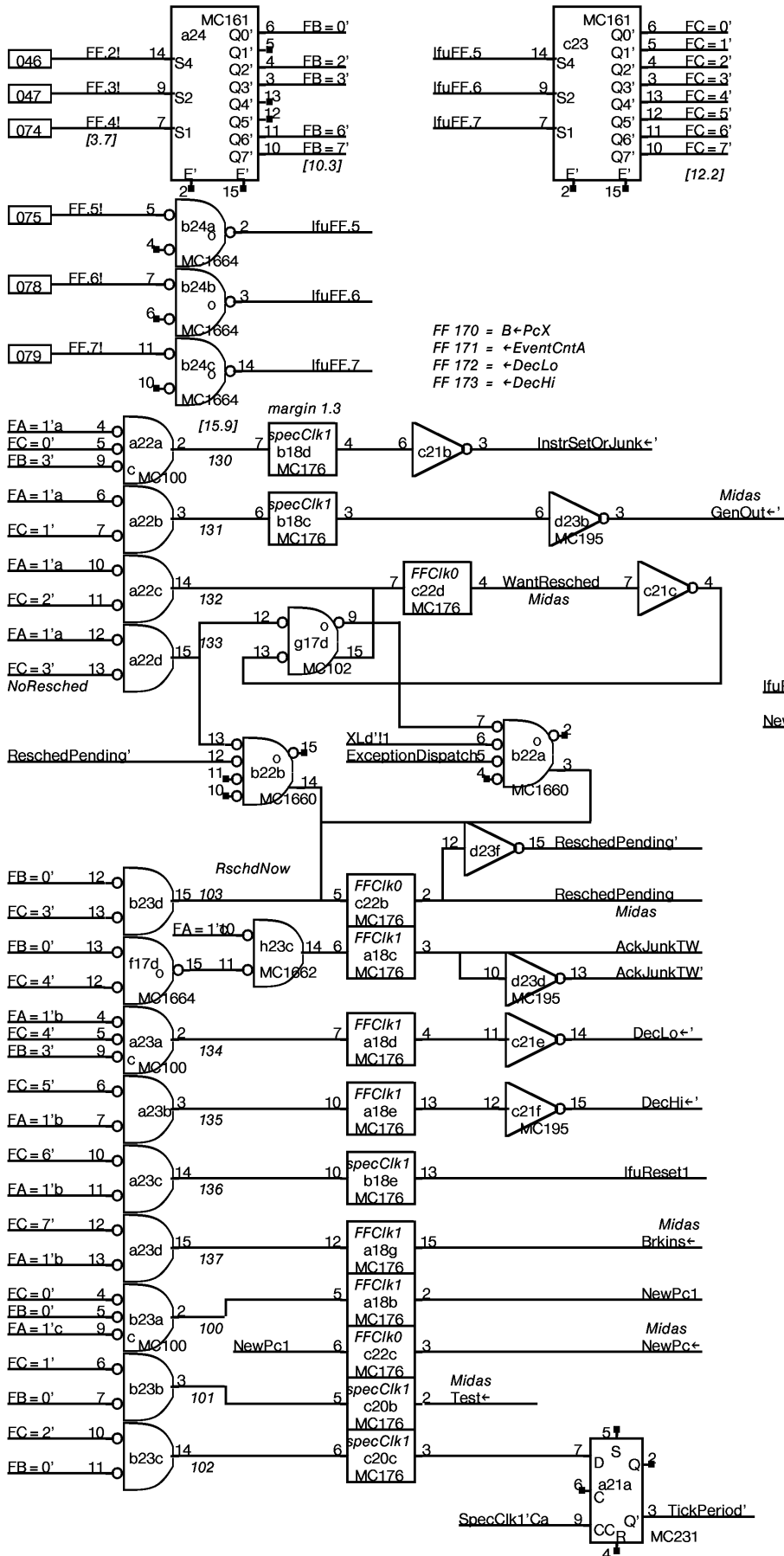
Instruction Fetch Unit

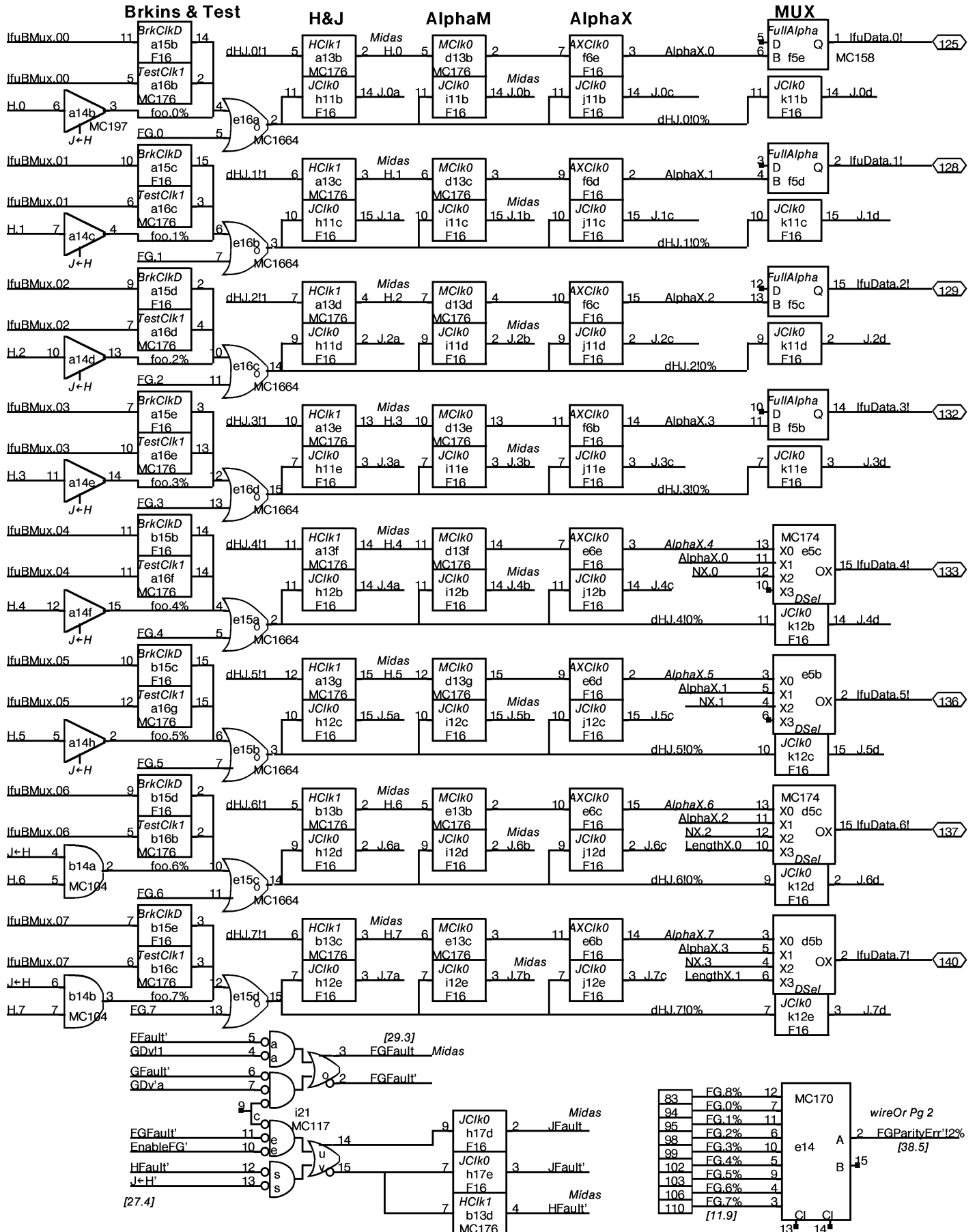
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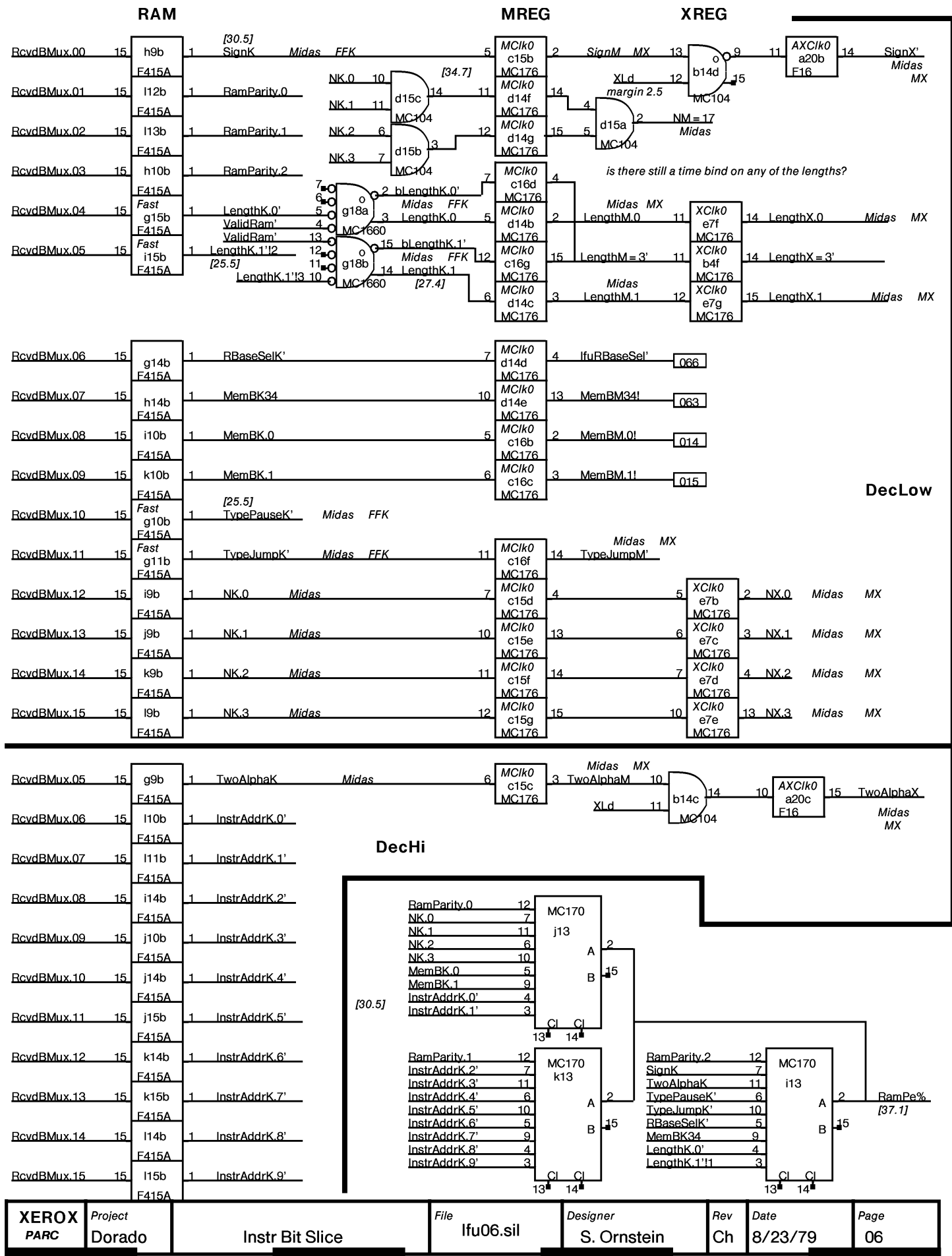
<u>TITLE</u>	<u>Page</u>
Mememory Request Logic _____	01
IFData, Jumps, and Exceptions _____	02
Loads and DVs _____	03
FF Decodes _____	04
Byte Bit Sclice _____	05
Instruction Bit Slice _____	06
Instruction Address Bit Slice _____	07
PC Bit Slice (High byte) _____	08
PC Bit Slice (Low byte) _____	09
Pc Pipe Control _____	10
Chip Control _____	11
Ifu Ram Control _____	12
Event Counter A _____	13
Event Counter B _____	14
Wakeups/Temperature Sensor _____	15
Midas control and multiplexors _____	16
Midas control and multiplexors _____	17
Clock distribution _____	18
Layout Upper Left quadrant _____	19
Layout Upper Right quadrant _____	20
Layout Lower Left quadrant _____	21
Layout Lower Right quadrant _____	22
IF Data Selector Prom _____	23



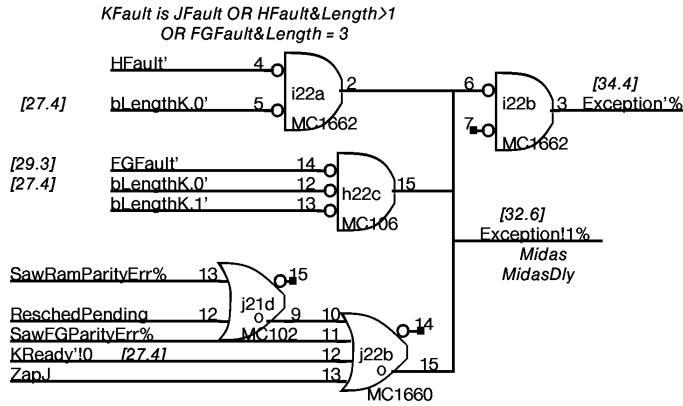




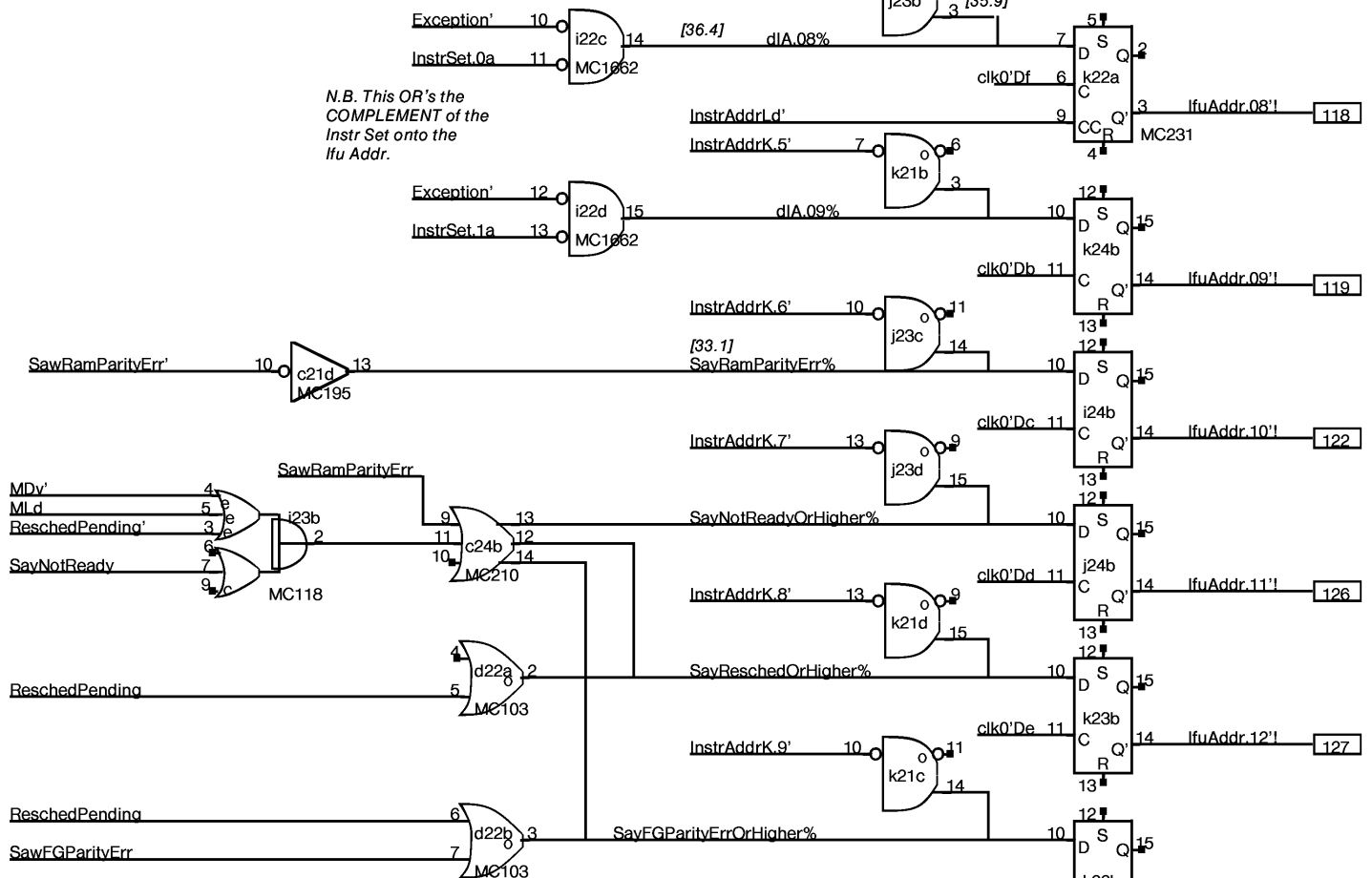




0 K Fault *low priority*
 4 FG Parity
 14 Resched
 34 Not Ready
 74 Ram Parity *high priority*

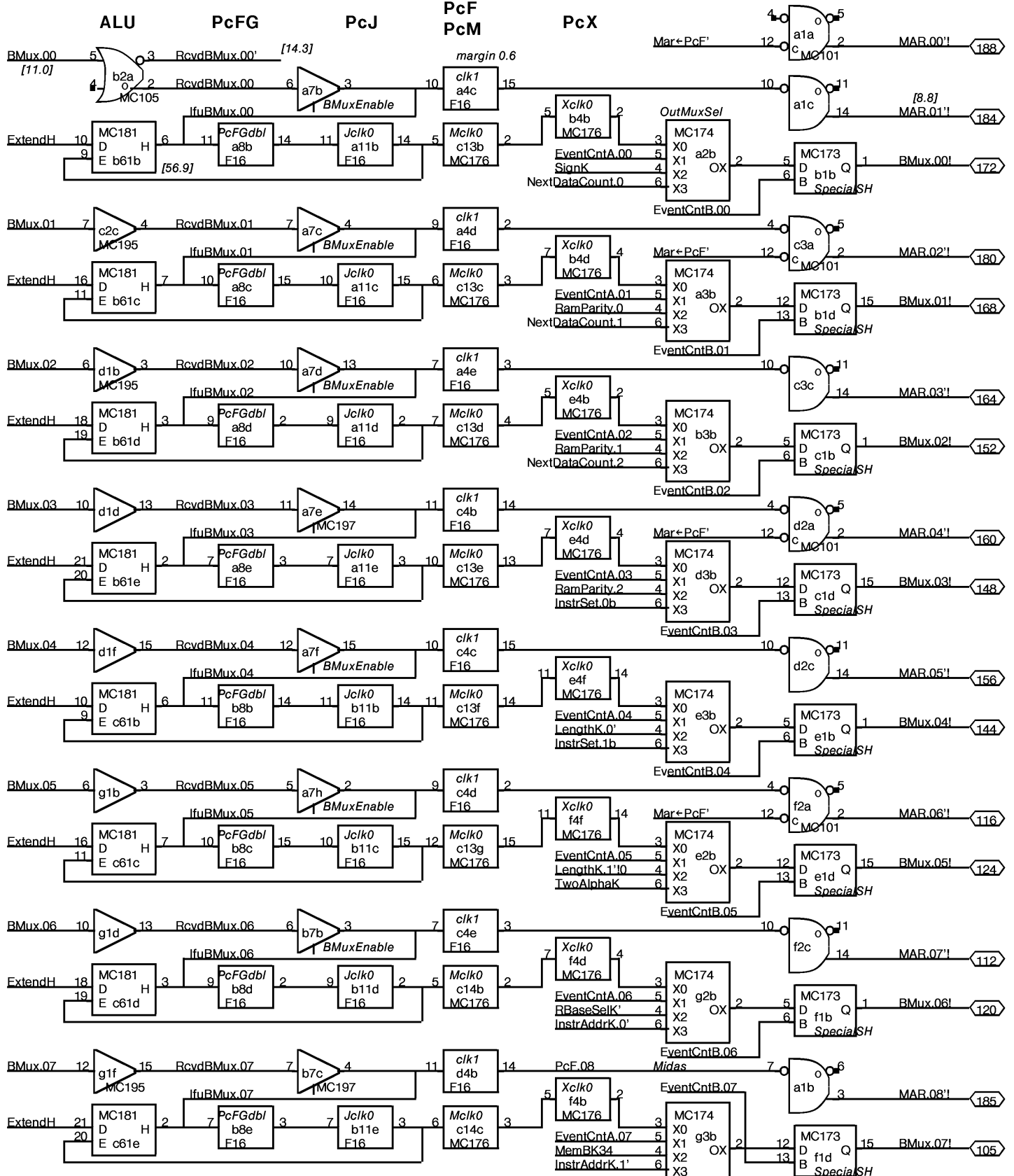


N.B. This OR's the
 COMPLEMENT of the
 Instr Set onto the
 Ifu Addr.



THE HIGH ORDER BYTE OF THE PC PIPE

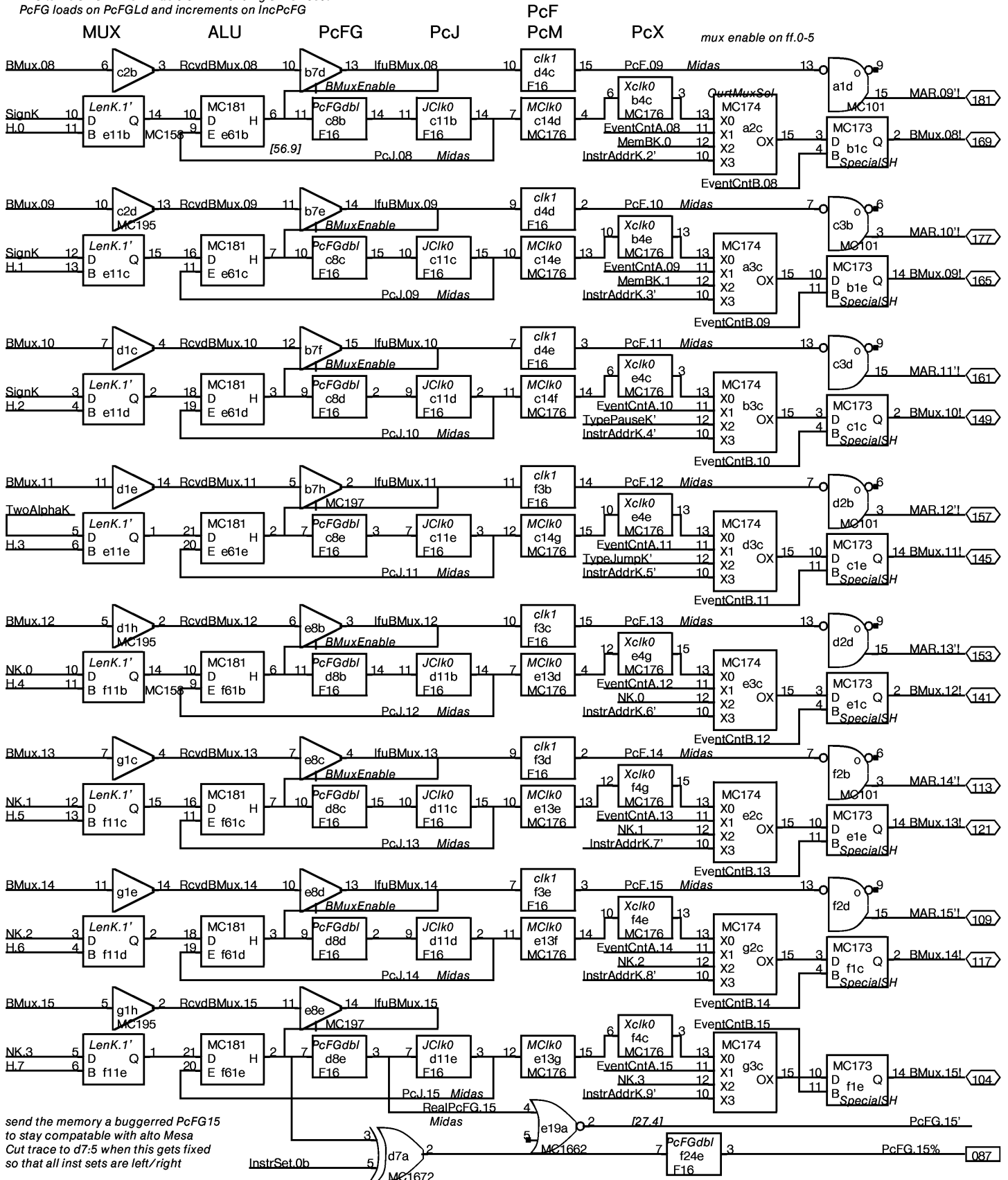
PcJ loads on JLD And not J+H, increments on JLD
 PcF Loads on ZapFGH and increments on IncPcF
 ALU turns on BMuxEnable or BrkPending or IfuReset1
 PcFG loads on PcFGLd and increments on IncPcFG



XEROX	Project	File	Designer	Rev	Date	Page
PARC	Dorado	PC Bit Slice (High byte)	lfu08.sil	S. Ornstein	Ch	8/23/79
						08

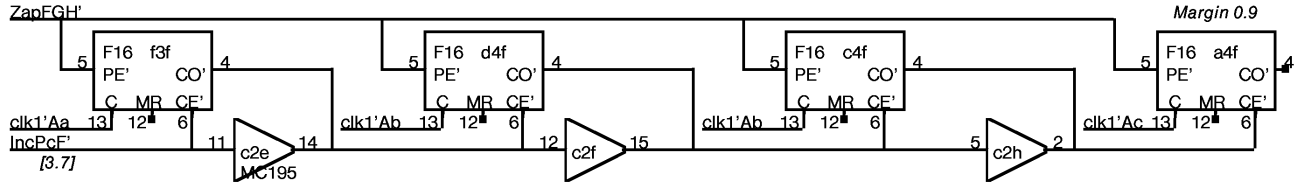
THE LOW ORDER BYTE OF THE PC PIPE

PcJ loads on Jld and not J+H, increments on Jld
PcF loads on ZapFGH and increments on IncPcF
ALU turns off on BMuxEnable or BrkPending or IfuReset
PcFG loads on PcFGLd and increments on IncPcFG



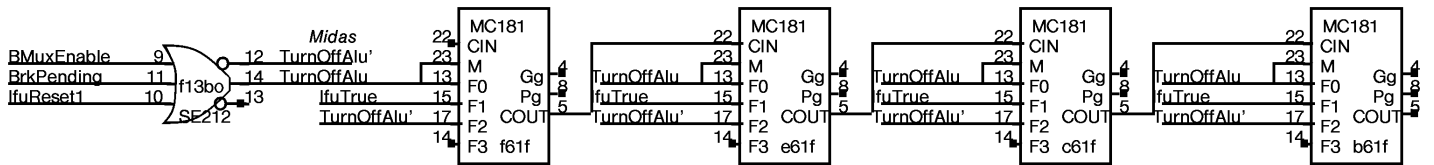
XEROX PARC	Project Dorado	PC Bit Slice (low byte)	File ifu09.sil	Designer S. Ornstein	Rev Ch	Date 8/14/79	Page 09
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PcF Control



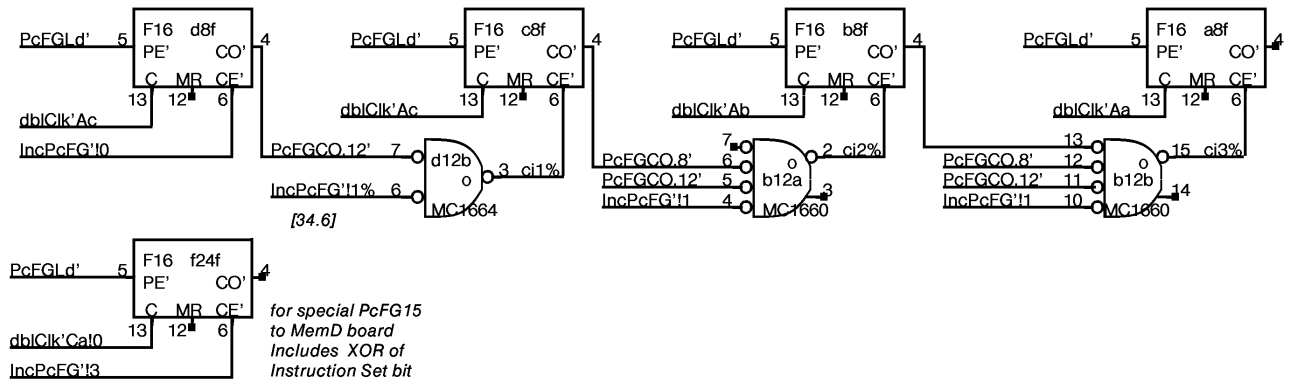
ALU Control

Low data in to high data out = 26.4



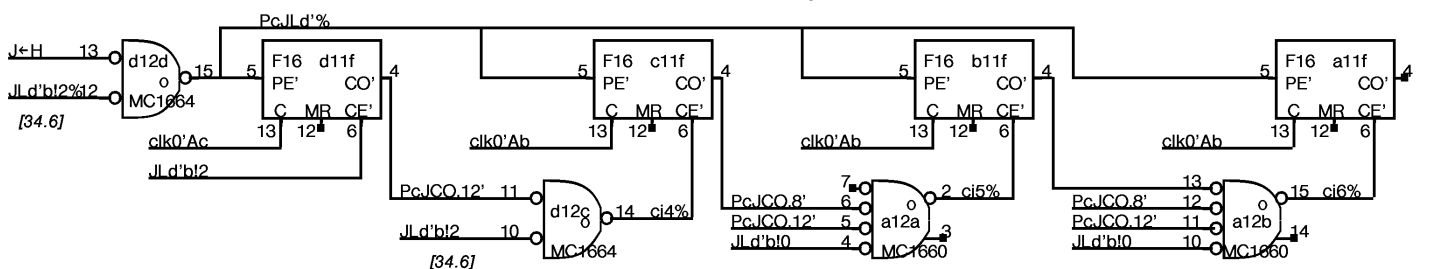
PcFG Control

[Margin 0.7 via Inc]



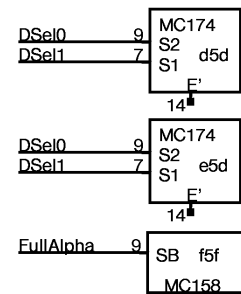
PcJ Control

[Margin 0.7 via load and Inc]

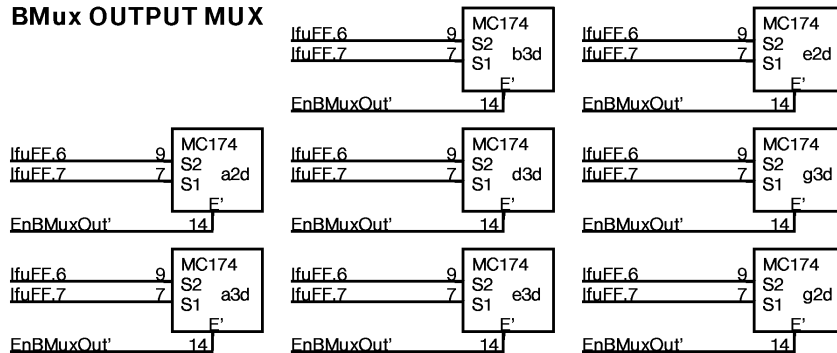


XEROX PARC	Project Dorado	Pc Control	File ifu10.sil	Designer S. Ornstein	Rev Ch	Date 8/23/79	Page 10
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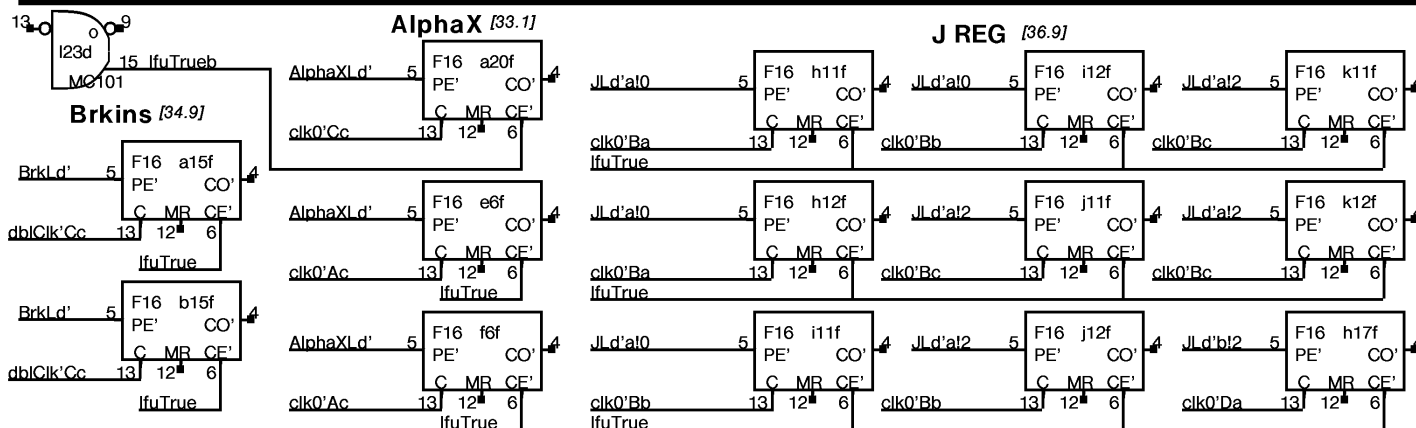
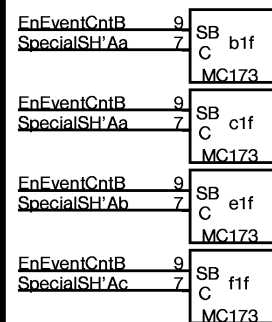
IfuData OUTPUT MUX



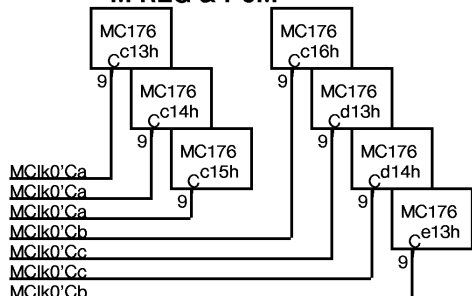
BMux OUTPUT MUX



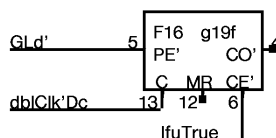
BMux Output Latches



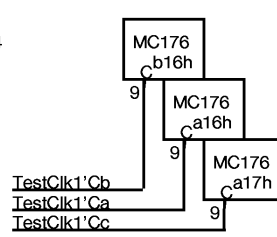
M REG & PcM [30.3]



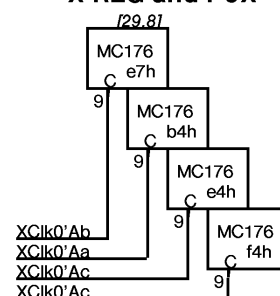
GFAULT & GDV



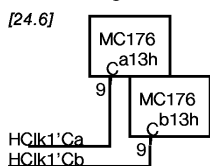
TEST REG



X REG and PcX [29.8]



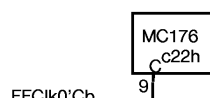
H REG



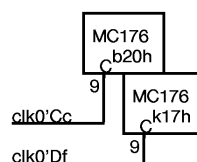
FF CLOCK 1



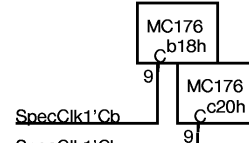
FF CLOCK 0



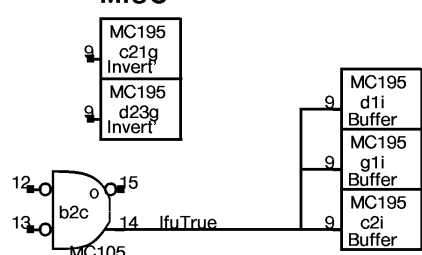
CLOCK 0



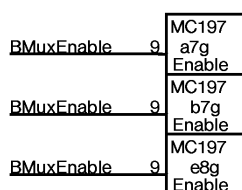
CLOCK 1 Without Test Mode



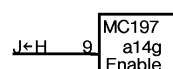
MISC



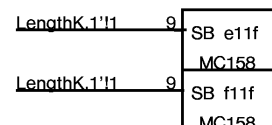
Use BMUX



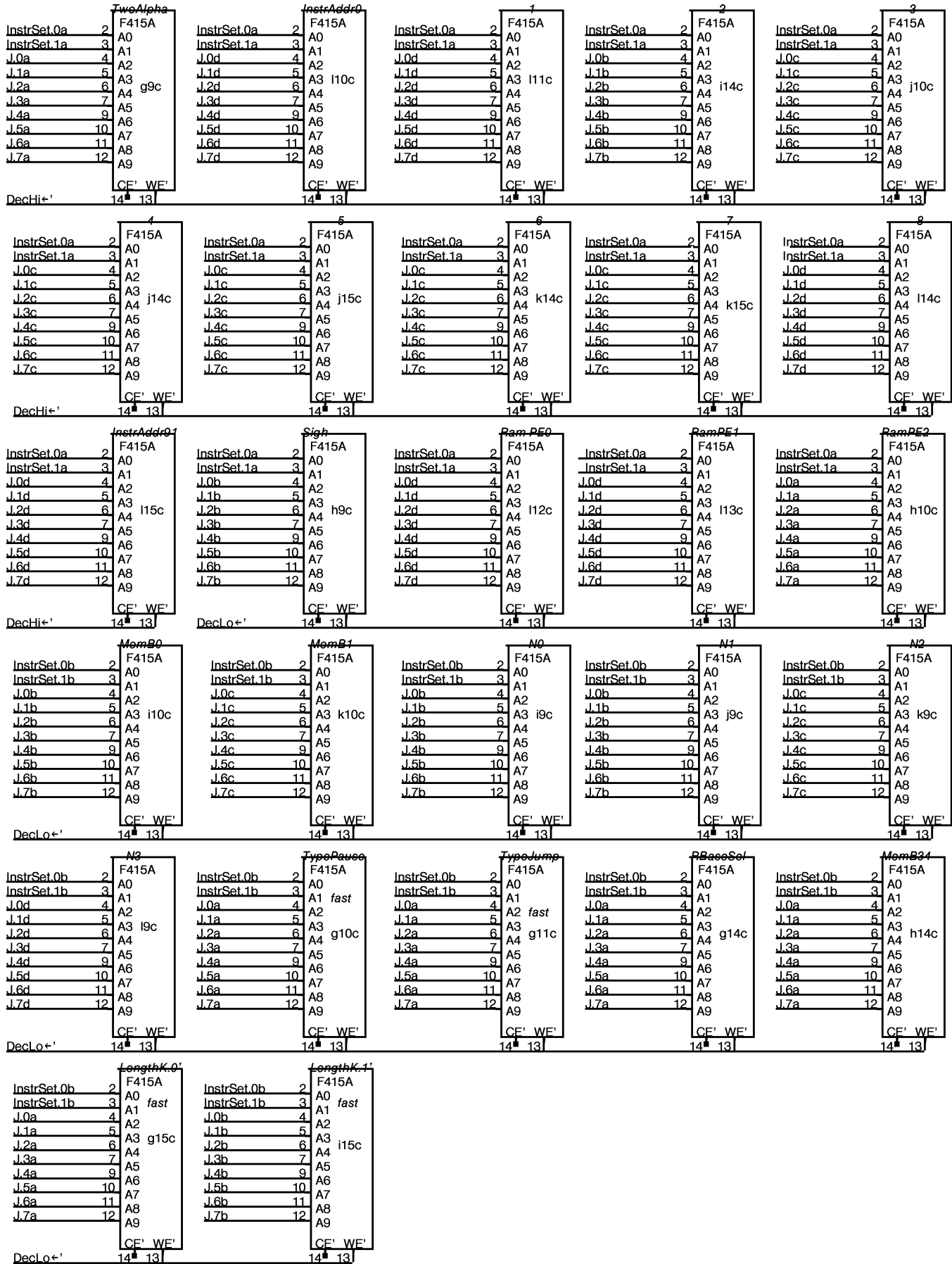
J Loads from H

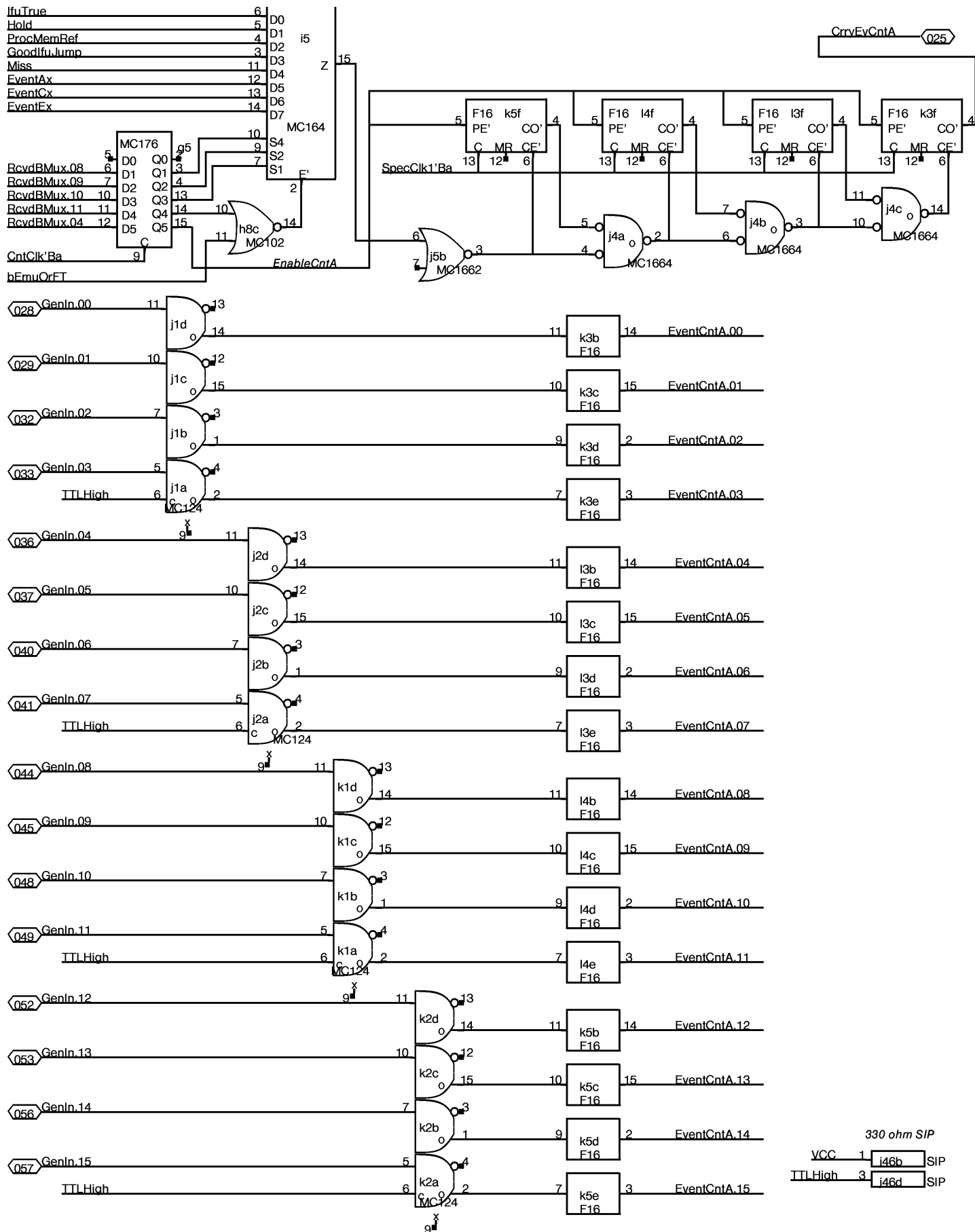


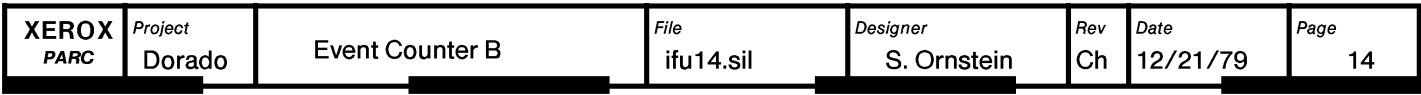
ALU INPUT MUX

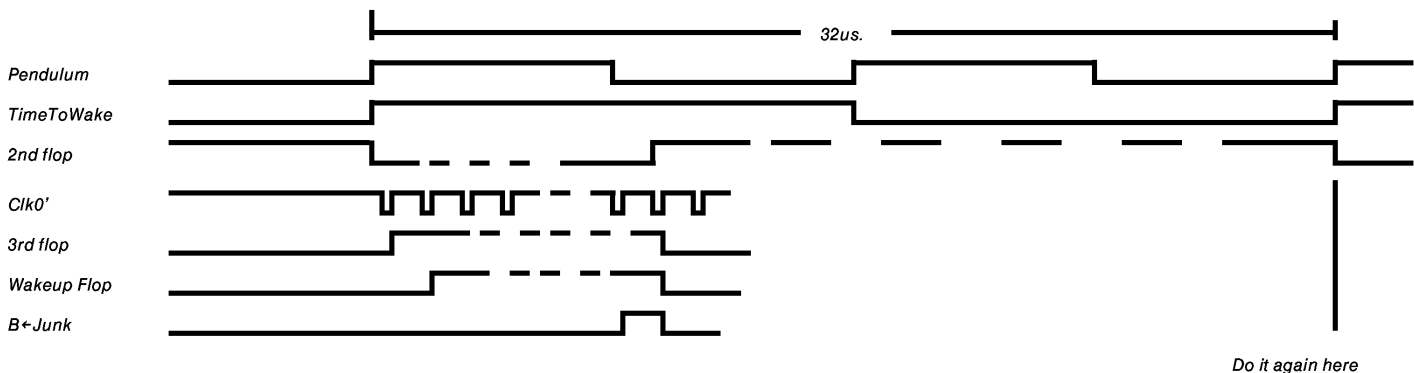
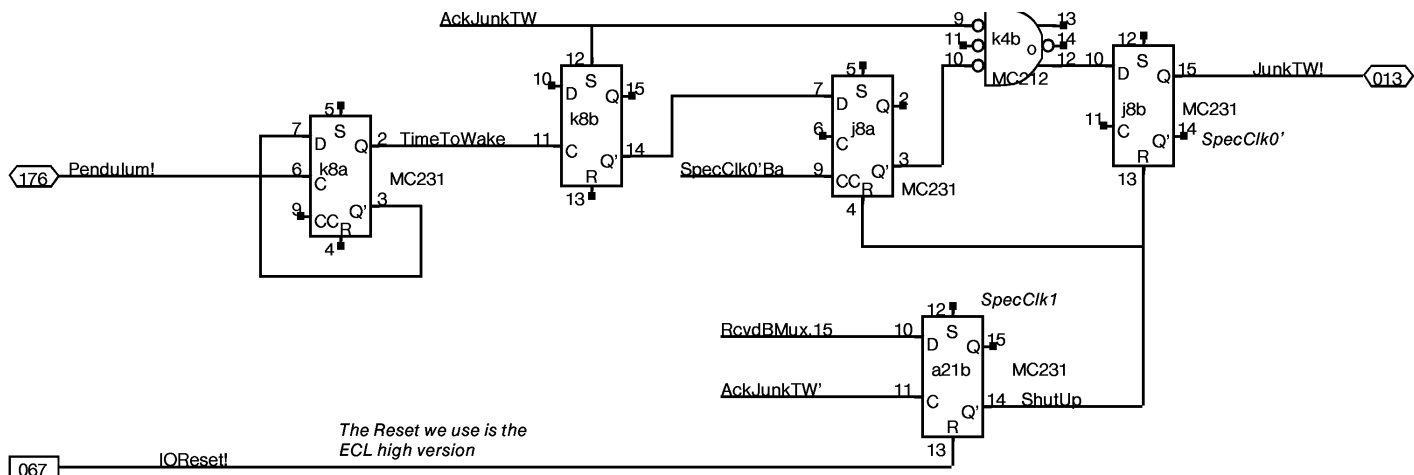


low order bits must
be fast!

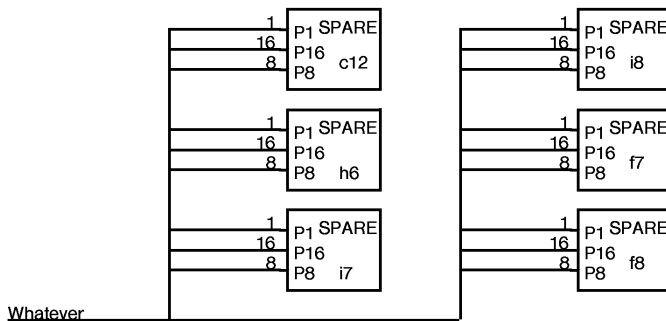
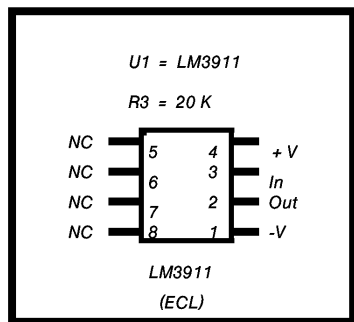
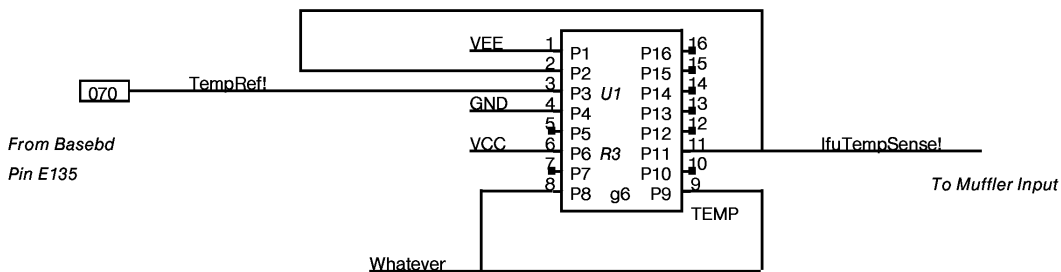




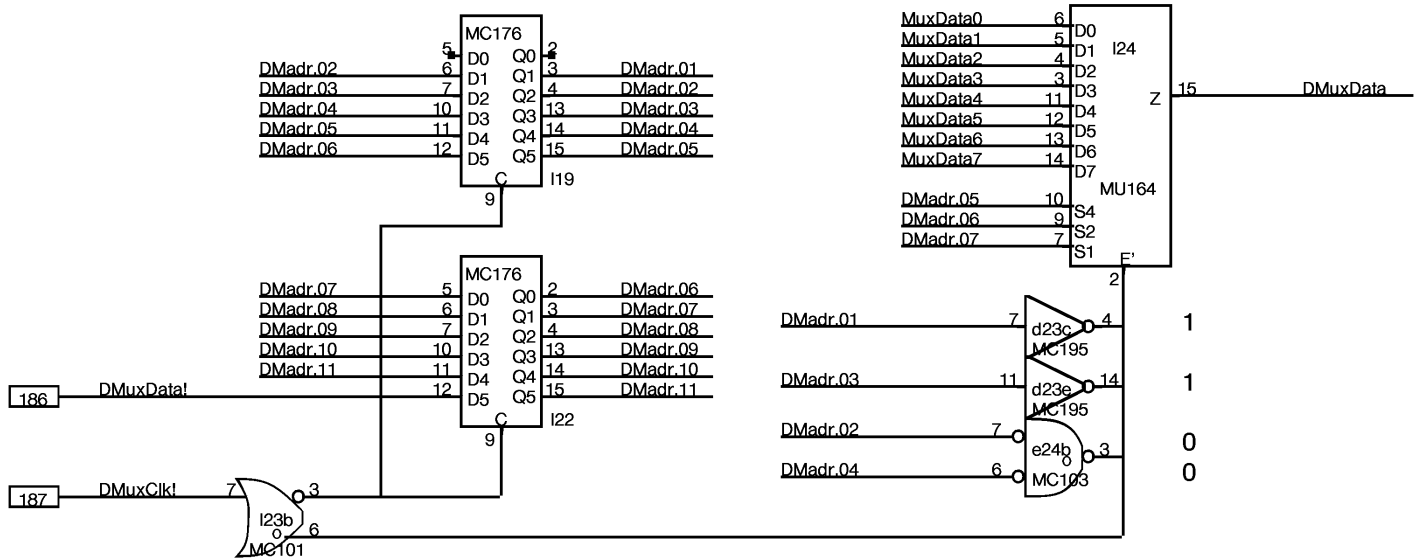




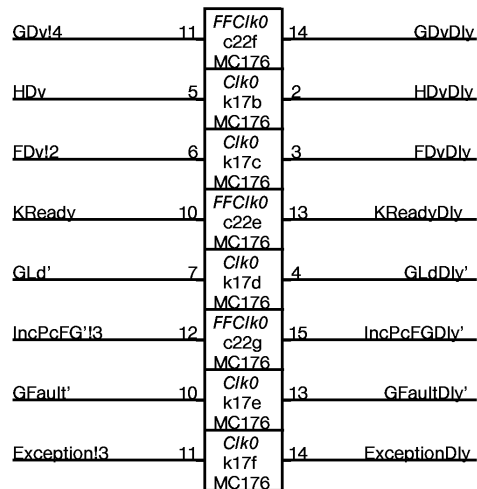
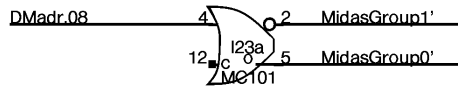
Temperature Sensor

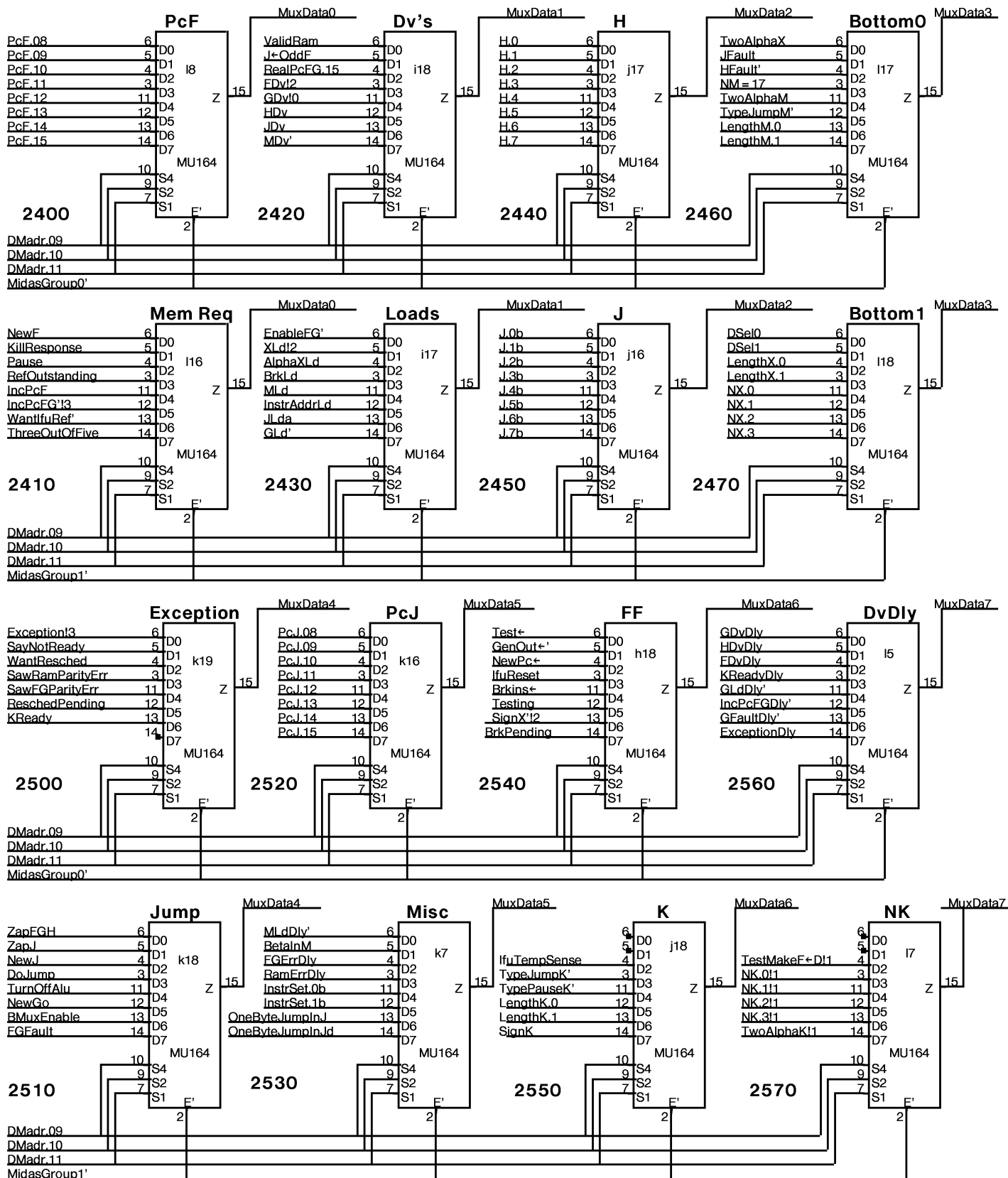


Muffler Control



We respond to 2400 thru 2577
Tho we are allocated to 2777





MAR 0,8,1,9,2,10

GenIn (28-57)

GenOut (60-89)

MAR 3,11,4,12,5,13

IfuData

BMux 0,8,1,9

BMux 2,10,3,11,4,12

BMux 5,13,6,14

BMux 7,15

01	A MAR 00 b MAR 08 c MAR 01 d MAR 09 101	b BMux 00 SS c BMux 08 d BMux 01 e BMux 09 173	b BMux 02 SS c BMux 10 d BMux 03 e BMux 11 173	b RcvdBmux 02 c RcvdBmux 10 d RcvdBmux 03 e RcvdBmux 11 f RcvdBmux 04 h RcvdBmux 12 195	b BMux 04 SS c BMux 12 d BMux 05 e BMux 13 173	b BMux 06 SS c BMux 14 d BMux 07 e BMux 15 173
02	b dBmux 00 c dBmux 08 174	a RcvdBmux 00 B FullAlpha c IfuTrue 105	b RcvdBmux 08 c RcvdBmux 01 d RcvdBmux 09 e PcFCarry f PcFCarry h PcFCarry 195	A MAR 04 b MAR 12 c MAR 05 d MAR 13 101	b dBmux 05 c dBmux 13 174	A MAR 06 b MAR 14 c MAR 07 d MAR 15 101
03	b dBmux 01 c dBmux 09 174	b dBmux 02 c dBmux 10 174	A MAR 02 b MAR 10 c MAR 03 d MAR 11 101	b dBmux 03 c dBmux 11 174	b dBmux 04 c dBmux 12 174	b PcF 12 c PcF 13 d PcF 14 e PcF 15 F16
04	b PcF 01 c PcF 02 d PcF 03 F16	b PcX 00 c PcX 08 d PcX 01 e PcX 09 f LengthX=3' g 176	b PcF 04 c PcF 05 d PcF 06 e PcF 07 F16	b PcF 08 c PcF 09 d PcF 10 e PcF 11 F16	b PcX 02 c PcX 10 d PcX 03 e PcX 11 f PcX 04 g PcX 12 176	b PcX 07 c PcX 15 d PcX 06 e PcX 14 f PcX 05 g PcX 13 176
05	b NextData 0 c NextData 1 d NextData 2 e F16	DSEL0 C0 141	A EnEventCntB b SignifuData c AlphaX' 106	b IfuData 7 c IfuData 6 174	b IfuData 5 c IfuData 4 174	b IfuData 3 c IfuData 2 d IfuData 1 e IfuData 0 158
06	DSELProm 139	DSEL1 C0 141	a clk1' A b 210	a SpecSH A b XClk0 A 210	b AlphaX 07 c AlphaX 06 d AlphaX 05 e AlphaX 04 F16	b AlphaX 03 c AlphaX 02 d AlphaX 01 e AlphaX 00 F16
07	b IfuBMux 00 c IfuBMux 01 d IfuBMux 02 e IfuBMux 03 f IfuBMux 04 h IfuBMux 05 197	b IfuBMux 06 c IfuBMux 07 d IfuBMux 08 e IfuBMux 09 f IfuBMux 10 h IfuBMux 11 197	a clk0' A b dbclk A 210	a PcFG.15 b dSaw FGPE c 1672	b NX.0 c NX.1 d NX.2 e NX.3 f LengthX.0 g LengthX.1 176	
08	b PcFG 00 c PcFG 01 d PcFG 02 e PcFG 03 F16	b PcFG 04 c PcFG 05 d PcFG 06 e PcFG 07 F16	b PcFG 08 c PcFG 09 d PcFG 10 e PcFG 11 F16	b PcFG 12 c PcFG 13 d PcFG 14 e PcFG 15 F16	b IfuBMux 12 c IfuBMux 13 d IfuBMux 14 e IfuBMux 15 f h 197	
09		b ALU 00 c ALU 01 d ALU 02 e ALU 03 181	b ALU 04 c ALU 05 d ALU 06 e ALU 07 181		b ALU 08 c ALU 09 d ALU 10 e ALU 11 181	b ALU 12 c ALU 13 d ALU 14 e ALU 15 181
10						
11	b PcJ 00 c PcJ 01 d PcJ 02 e PcJ 03 F16	b PcJ 04 c PcJ 05 d PcJ 06 e PcJ 07 F16	b PcJ 08 c PcJ 09 d PcJ 10 e PcJ 11 F16	b PcJ 12 c PcJ 13 d PcJ 14 e PcJ 15 F16	b dALU 08 c dALU 09 d dALU 10 e dALU 11 158	b dALU 12 c dALU 13 d dALU 14 e dALU 15 158
12	a PcJCarry b PcJCarry 1660	a PcFGCarry b PcFGCarry 1660		a dIncPcFG b PcFGCarry c PcJCarry d PcJLd 1664	a FH b SH 210	a preClk1 b preClk1 210

A

B

C

D

E

F

XEROX
PARCProject
Doradoreference
Upper Left QuadrantFile
ifu19.silDesigner
S. OrnsteinRev
ChDate
8/08/80Page
19

		102		84		68		52		36		20		x RunRfsh x CLK.ifu x ClkEnable x StartCycle	
01	b RcvdBmux 05 c RcvdBmux 13 d RcvdBmux 06 e RcvdBmux 14 f RcvdBmux 07 h RcvdBmux 15	195	A GenOut.11 b GenOut.10 c GenOut.09 d GenOut.08	125	A GenOut.07 b GenOut.06 c GenOut.05 d GenOut.04	125	A Genin.03 b Genin.02 c Genin.01 d Genin.00	124	A Genin.11 b Genin.10 c Genin.09 d Genin.08	124	A prepreFH'a b See Pg. 2	231			
	b dBmux 06 c dBmux 14	174	A GenOut.15 b GenOut.14 c GenOut.13 d GenOut.12	125	A GenOut.03 b GenOut.02 c GenOut.01 d GenOut.00	125	A Genin.07 b Genin.06 c Genin.05 d Genin.04	124	A Genin.150 b Genin.14 c Genin.13 d Genin.12	124	a ClkEnable b prepreDblClk	210			
03	b dBmux 07 c dBmux 15	174	Spec 1 EventCntB F16	Spec 1 EventCntB F16	a b BCntCarry b BCntCarry c BCntCarry d BCntCarry	1664	Spec 1 EventCntA F16	Spec 1 EventCntA F16							
04	CntClk BCntMuxCntrl	176	Spec 1 EventCntB F16	Spec 1 EventCntB F16	a ACntCarry b ACntCarry c ACntCarry d	1664	a b dTWReq01	212	Spec 1 EventCntA F16						
05	CntClk ACntMuxCntrl	176	BCntMux 164	ACntMux 164	a b dCntA b dCntB c dCntB d dCntB	1662	Spec 1 EventCntA F16	DvDIy MU164							
06	Temp Sensor			a CntClk B b SpecClk1 B	210	A IfuReset b NewGo SpecClk0	231	b bEmuOrFT c Hold d ProcMemRef e IfuMemRef f GoodIfuJump g EventAx	Spec 1 176	b EventBx c EventCx d EventDx e EventEx f Miss g NotRdyJmp	Spec 1 176				
07	c0 Events	176	c0 Events	176		a clk0 B b SpecClk0 B	210	Misc MU164	NK MU164						
08	a EventCntB b EventCntB c EventCntB D EventCntB	104	a BCntMuxEnb b blfuHold' c ACntMuxEnb D bEmuOrFT	102		SpecClk0 A SyncedTTW b TWReq01	231	A TimeToWake b del TTW Pendulum	231	PcF MU164					
09	TwoAlphaK	415	SignK	415	NK.0	415	NK.1	415	NK.2	415	NK.3	415			
10	TypePauseK	415	Parity2	415	MEMBK.0	415	IfuAddr 3	415	MemBK.1	415	IfuAddr 0	415			
11	TypeJumpK	415	b J0a c J1a d J2a e J3a	c0 F16	b J0b c J1b d J2b e J3b	c0 F16	b J0c c J1c d J2c e J3c	c0 F16	b J0d c J1d d J2d e J3d	c0 F16	IfuAddr 1	415			
12	a preClk0 b preClk0	210	b J4a c J5a d J6a e J7a	c0 F16	b J4b c J5b d J6b e J7b	c0 F16	b J4c c J5c d J6c e J7c	c0 F16	b J4d c J5d d J6d e J7d	c0 F16	Parity0	415			
G		H		I		J		K		L					

13	b H 0 c H 1 d H 2 e H 3 f H 4 g H 5 176	b H.6 c H.7 d HFault e f g 176	b PcM 00 c PcM 01 d PcM 02 e PcM 03 f PcM 04 g PcM 05 176	b AlphaM 0 c AlphaM 1 d AlphaM 2 e AlphaM 3 f AlphaM 4 g AlphaM 5 176	b AlphaM 6 c AlphaM 7 d PcM 12 e PcM 13 f PcM 14 g PcM 15 176	a preFH b TurnOffAlu 212
14	b EnableH 0 c EnableH 1 d EnableH 2 e EnableH 3 f EnableH 4 g EnableH 5 197	a EnableH 6 b EnableH 7 c dSignX d TwoAlphaX 104	b PcM 06 c PcM 07 d PcM 08 e PcM 09 f PcM 10 g PcM 11 176	b LengthM.0 c LengthM.1 d SpareM.0 e SpareM.1 f dNM = 17a g dNM = 17b 176	FGParityErr 170	b InstrSet 0a c InstrSet 0b d InstrSet 1a e InstrSet 1b FFc1 F16
15	b Brkins 0 c Brkins 1 d Brkins 2 e Brkins 3 F16	b Brkins 4 c Brkins 5 d Brkins 6 e Brkins 7 F16	b SignM c TwoAlphaM d NM 0 e NM 1 f NM 2 g NM 3 176	a NM = 17 b NK2&NK3 c NK0&NK1 d IfuMemAck 104	a dHJ4 b dHJ5 c dHJ6 d dHJ7 1664	a InstrSetLd b CntLd c d TestOrReset 103
16	b TestReg 0 c TestReg 1 d TestReg 2 e TestReg 3 f TestReg 4 g TestReg 5 176	b TestReg 6 c TestReg 7 d e TestFH f TestSH g Testing 176	b MemBM 0 c MemBM 1 d LengthM = 3 e MDv f TypeJumpM g LengthM = 3 176	A BrkPending b 231	a dHJ0 b dHJ1 c dHJ2 d dHJ3 1664	IncPcFG 119
17	b TestIfuAck c TestMemAck d TestMakeF+D e TestFault f g 176	PipeClkEn 121	A SawFGPe b (delayed clock) 231	b AlphaXLd c XShift 117	A Pause b IncPcF c0 231	a dPause b PcFGLd c blfuHold d dAckJunkTW 1664
18	b NewPc1 c AckJnkTW d DecLo+ e DecHi+ f g Brkins+ 176	b GenOut+ c InstrSetOrJunk+ d IfuReset1 e f g Spec1 176	a HClk1' b TestClk1' C C 210	a MCik0' b FFCik1' C C 210	a XLd b KReady 1660	InstrAddrLd 121
19	a BMuxEnable B NextBeta 109	a dNewGo B EnBMuxOut c TestLd 105	a dbiclk' b SpecClk1' C C 210	a FFCik0' b Clk0' C C 210	a PcFG15 b KReady c Kready d dBrkLd 1662	a JKilResponse b FGdV 212
20	b SignX c TwoAlphaX d BetaInM e F16	b blfuFault c NewF d NewF' e MLdDly f DoJump g DoJump' 176	b Test+ c dTickPeriod d RamErrDly e FGErrDly f WantIfuHoldDly g 176	b ExtendH c slncPcF 117	a B dSawFGPe c InstrAddrLd 105	A HDv b FDv c2 135
21	A TickPeriod b ShutUp (junk) Spec1 231	b c WantIfuHold 117	b InstrSetOrJunk+ c ReschedPending d SayRamParityErr e DecLo+ f DecHi+ g TestMemAck' inv 195	b dJDv c ZapJ 117	b BrkLd c SayNotReady 117	A NewJ b RefOutstanding c0 135
22	A FF130 b FF131 c FF132 d FF133 100	a dReschedPending b dReschedPending 1660	b ReschedPending c NewPc+ d WantResched e KReadyDly f GDvDly g IncPcFGDly 176	a SayResched b SayFGParityErr c BetaInM' d IfuNextData 103	a BetaInH B JFDv c NewGoOrNoM 105	a dSayNotReady b K NewJ c dGDv d KFDv 1664
23	A FF134 b FF135 c FF136 d FF137 100	A FF100 b FF101 c FF102 d dReschedPending 100	FC 161	b GenOut+ c DMuxEnable d AckJnkTW' e DMuxEnable f ReschedPending' g SayNotReady inv 195	a ZapFGH B JHDv c SeeJump 105	a EnableFG' b J+H 1660
24	FB 161	a FF5 b FF6 c FF7 d EmuOrFT' 1664	a FA = 1 b SayStuff 210	a J+OddF b SayNotReady1 1660	a EnableFG' b DMuxEnable C dMiss d 1ByteJumpInJ 103	b c d e specialPcFG15 c2 F16

A 10 B 26 C 42 D 58 E 74 F 92

MemB
x EmuOrFT' x NextData
x WantIfuHold
xASEL0

FF FG

XEROX	Project	reference	File	Designer	Rev	Date	Page
PARC	Dorado	Lower Left Quadrent	ifu21.sil	S. Ornstein	Ch	12/21/79	21

13	a preDbIClk b PipeClkEn 211	A SawRamParityErr b c0 231	RamParityErr 2 170	RamParityErr 0 170	RamParityErr 1 170	Parity1 415
14	RBaseSel 415	MemB 34 415	IfuAddr 2 415	IfuAddr 4 415	IfuAddr 6 415	IfuAddr 8 415
15	LengthK.0' 415	a HDv b HDv c KReady d dGLd 1662	LengthK.1' 415	IfuAddr 5 415	IfuAddr 7 415	IfuAddr 9 415
16	a MightBeJump B WantIfuRef c IfuMakeF+D 105	a d HDv b ZapOrJEmpty c dJDv D dSawFGPE 102	A ExceptionDispatch b 231	J MU164	PcJ MU164	Top MU164
17	a dPause b c RealJFault D dWantResched 102	b JDv' c JDv d JFault e JFault' c0 F16	Loads MU164	H MU164	b HDvDly c0 c FDvDly d GLdDly' e GFaultDly' f ExceptionDly' g 176	Bottom MU164
18	a LengthK.0 b LengthK.1 1660	FF's MU164	Dv's MU164	K MU164	Jumps MU164	Bottom MU164
19	b GDv'a c GDv' d GFault e GDv c2 F16	OtherHole 121	a clk0' D b clk0' D 210	a dbclk' D b clk1' D 210	Exception MU164	DMadr1-5 176
20	dGDv 121	WantIfuRef2 121	A 1ByteJumpInJ b 1ByteJumpInJd c0 135	A Mar+PcF b FFault c1 231	a WantIfuRef1 b WantIfuRef1 c WantIfuRef1 D 102	a K KillResponse b IfuAddr 6 C K RefOutstanding d IfuAddr 7 103
21	a WantIfuRef1 b WantIfuRef2 c WantIfuRef3 d WantIfuRef' 1662	dGDv 119	b FGFault c FGFault 117	a K HDv b K HDv c K HDv D Exception 102	A IfuAddr 4 b IfuAddr 9 c IfuAddr 53 d IfuAddr 12 101	A KillResponse b c2 135
22	GLd 119	A dGLd1 b dGLd2 c Exception 106	a RealHFault b Exception' c IfuAddr 8 d IfuAddr 9 1662	a ValidRam B Exception 1660	A IfuAddr 4 c0 b IfuAddr 9 c0 231	DMadr6-11 176
23	JLdb 121	a bMLd b dMiss c dAckJunkTW d 1662	b SayNotReady c dMiss 118	A IfuAddr 5 b IfuAddr 8 c IfuAddr 10 d IfuAddr 11 101	A IfuAddr 3 c0 b IfuAddr 8 c0 231	A MidasGroup b DMuxClk c d IfuTrueb 101
24	MLd 121	JLda 121	A IfuAddr 1 c0 b IfuAddr 6 c0 231	A IfuAddr 2 c0 b IfuAddr 7 c0 231	A IfuAddr 0 c0 b IfuAddr 5 c0 231	DMuxData MU164

92 **G** 110 **H** 126 **I** 142 **J** 158 **K** 174 **L**

FG0-6 IfuAddr0-9 x IfuNextMacro x WantIfuRef MakeF+D IfuAck DMuxData DMuxClock

IF Data Selector PROM SG 139 (see Dwg. 02)

N	Two Alpha	JMP	Length	Address	Contents	Numeric Sequence	Effective Sequence
N		 	0	00	11111111	3, 3, 3, 3, 3	IL, IL, IL, IL, IL.....
			1	01	11111111		
			2	02	11111111		
			3	03	11111111		
			0	04	11111111	2, 3, 3, 3, 3	N, IL, IL, IL, IL.....
			1	05	11110111		
			2	06	10110011		
	 Two Alpha 	 	3	07	10010001	3, 3, 3, 3, 3	N, Alpha, IL, IL, IL..... N, Alpha, Beta, IL, IL.....
			0	10	11111111		
			1	11	11111111		
			2	12	11111111		
			3	13	11111111	2, 1, 0, 3, 3	N, Alpha[0:3], Alpha[4:7], IL, IL..... N, Alpha[0:3], Alpha[4:7], Beta, IL.....
			0	14	11111111		
			1	15	11111111		
no N		 	2	16	10010101	3, 3, 3, 3, 3	IL, IL, IL, IL, IL.....
			3	17	10001000		
			0	20	11111111		
			1	21	11111111		
			2	22	11111111	0, 3, 3, 3, 3	Alpha, IL, IL, IL, IL..... Alpha, Beta, IL, IL, IL.....
			3	23	11111111		
			0	24	11111111		
	 Two Alpha 	 	1	25	11111111	3, 3, 3, 3, 3	IL, IL, IL, IL, IL.....
			2	26	01110111		
			3	27	00110011		
			0	30	11111111		
			1	31	11111111	1, 0, 3, 3, 3	Alpha[0:3], Alpha[4:7], IL, IL, IL..... Alpha[0:3], Alpha[4:7], Beta, IL, IL.....
			2	32	11111111		
			3	33	11111111		
			0	34	11111111	1, 0, 0, 3, 3	Alpha[0:3], Alpha[4:7], Beta, IL, IL.....
			1	35	11111111		
			2	36	00111011		
			3	37	00011001		

No zero length instructions
 No 3 - Byte Jumps
 Jumps never export anything other than the length (no "N" on 2 - byte Jumps).
 A one - byte instruction specifying Two Alpha is meaningless